

Electronic System Design: Begin With XSCHEM

SHANOSH KUMAR

The need for a schematic editor arises from the complex electronic systems that were once manually drawn by engineers and designers. Today, most modern schematic software are equipped with state-of-the-art tools that can handle very complex circuit conception to its final product. This is how digital schematics were born.

XSCHEM could be that tool of choice that lets you work with nets, pins and wires. Knowing these basics marks the beginning of formal circuit design followed by simulation. During the first step in a design cycle, you have to make sure that the tools in the schematic editor are well understood. Once the schematic has been drawn on the computer, circuit connectivity and device netlist are generated and sent to a circuit simulator.

Judging a book by its cover

To make this process effective while transcending the concept from your mind to the editor, an easy graphical interface is required. By focusing on interfaces, hierarchy and instance properties, this software helps simplify a complex system and creates an illustration that looks like small building blocks of the project.

XSCHEM is made more attractive by its capability to handle integrated circuit (IC) design and to generate netlists for very-large-scale digital, analogue or mixed-mode simulations.

A netlist is generated with every simulation, and that provides a textual description of a circuit made of components. Components such as gates, resistors, capacitors and

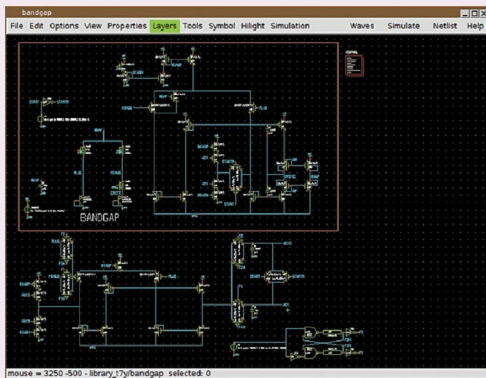


Fig. 1: Analogue circuit example

transistors can all be listed together to form a netlist when used in analogue simulation tools. Once the schematic is created, a circuit netlist can be generated for simulation. Currently, XSCHEM supports three netlist formats:

- SPICE netlist
- VHDL netlist
- VERILOG netlist

Give the benefit of the doubt

XSCHEM could be looked upon as a featureful industrial-grade software. As you can see in Fig. 1, it is a depiction of VLSI system(s)-on-chip imported in XSCHEM. In the figure, the tool is seen handling more than ten hierarchies, and the primary goal of this program is to build a bigger schematic. This has been the primary goal during the whole

development of the program.

Although the user interface looks very simple, the net-listing and rendering engines in XSCHEM are designed from the ground up to handle very large designs in the most efficient way. All you have to do is keep bind keys in your hands, as most of the work is done by these. This comes in very handy when working on bigger projects that combine more than 1000 transistors and still use context menus and elaborate graphical actions while designing these.

Start designing using layers

Most editing commands are available in the menu, but definitely key bindings and mouse actions are the most effective ways to build and arrange schematics, so you should